

Database_VValidation_Testing_Report

Test 1: Duplicate User Emails

Business Rule

User emails must be unique.

Query

```
SELECT email, COUNT(*) as count
FROM users
GROUP BY email
HAVING COUNT(*) > 1;
```

Result

email	count
john@gmail.com	2

Issue Found

Duplicate email addresses exist in the database.

Impact

- Can cause login issues
- Duplicate accounts for same user
- Incorrect user data tracking

Severity: Medium

Test 2: Negative Product Price

Business Rule

Product price must be greater than 0.

Query

```
SELECT *  
FROM products  
WHERE price < 0;
```

Result

id	name	price	stock	is_active
2	mouse	-20.0	80	1

Issue Found

Product has negative price.

Impact

- Financial loss
- Incorrect billing
- Data integrity issue

Severity: High

Test 3: Active Products with Zero Stock

Business Rule

Active products should not have zero stock.

Query

```
SELECT *  
FROM products  
WHERE is_active = 1 AND stock = 0;
```

Result

id	name	price	stock	is_active
3	Keyboard	100.0	0	1

Issue Found

Product is active but has zero stock.

Impact

- Poor user experience
- Customers cannot purchase item
- Loss of trust

Severity: Medium

Test 4: Orders with Invalid User

Business Rule

Every order must have a valid user.

Query

```
SELECT o.id as order_id, o.user_id
FROM orders o
LEFT JOIN users u ON o.user_id = u.id
WHERE u.id IS NULL;
```

Result

order_id	user_id
2	5

Issue Found

Order linked to a non-existent user.

Impact

- Order cannot be tracked
- Delivery issues
- Reporting errors

Severity: High

Test 5: Order Items with Invalid Product

Business Rule

Order items must reference valid products.

Query

```
SELECT oi.id as order_item_id, oi.product_id
FROM order_items oi
LEFT JOIN products p ON oi.product_id = p.id
WHERE p.id IS NULL;
```

Result

order_item_id	product_id
3	10

Issue Found

Order item references non-existing product.

Impact

- Order processing failure
- Incorrect billing
- Inventory mismatch

Severity: High